

Spider Seeder

ME 102B — Mechatronics Design · Spring 2026 **Team #21**

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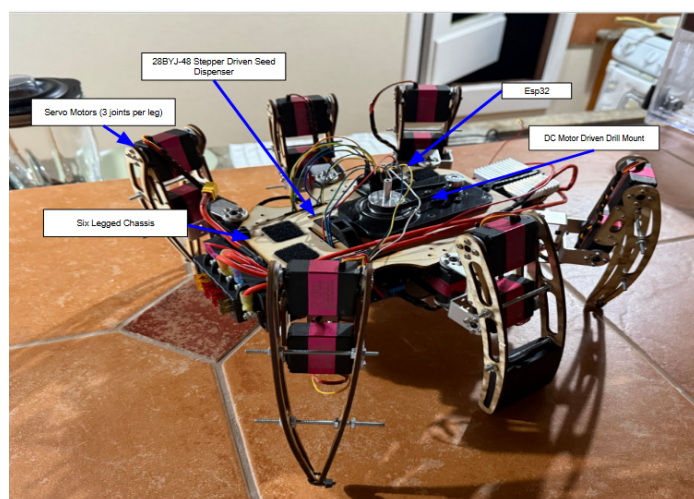
1. Opportunity

Ecological restoration requires planting large numbers of seeds in uneven, remote terrain where manual methods are labor intensive and conventional machinery lacks precision. The Spider Seeder Robot fills this gap with a compact, autonomous platform that navigates rough ground, drills to consistent depth, and dispenses seeds in a single integrated cycle using low-cost components suited for replication and future development.

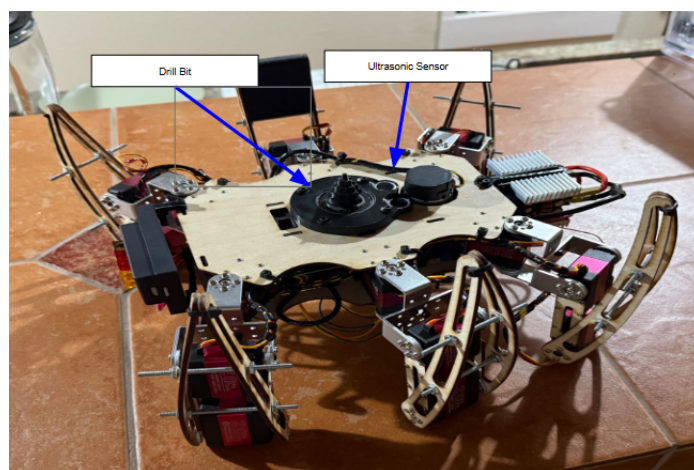
2. System Overview and High-Level Strategy

The robot integrates five subsystems: a six-legged servo-driven chassis, a DC motor drill, a stepper-motor seed dispenser, a sensor suite, and an ESP32 control and communication layer (Figure 1).

The robot has the ability to achieve locomotion; however, the robot plants from a fixed standing position using an event-driven state machine: it lowers its body via coxa servo deflection to drive the drill into soil, retracts, dispenses seeds, and sweeps a leg to cover the hole. This focused the control effort on depth sensing and actuation reliability over gait planning, appropriate for a first-generation prototype.



(a) Annotated top view.



(b) Annotated bottom view.

Figure 1: Annotated photos of the fully assembled Spider Seeder Robot. The labeled views identify the main mechanical, sensing, and actuation subsystems, including the six-legged chassis, servo motors, drill subsystem, ultrasonic sensor, ESP32 controller, and stepper-driven seed dispenser. These photos serve as a visual reference for the integrated prototype and the key components used during the planting sequence.

3. Desired vs. Achieved Specifications

Table 1 compares design targets against achieved performance. Values marked XX will be filled with measured results from final testing.

Table 1: Design targets vs. achieved specifications.

Parameter	Target	Achieved
Drilling depth	5 cm	7 cm
Drill motor actuation	Full PWM (12 V, 255/255)	Full PWM (12 V, 255/255)
Body descent range (coxa)	90 $\rightarrow$ 45	90 $\rightarrow$ 45
Depth sensing method	Tare + relative threshold	Implemented, ± 0.5 cm
Covering sweep reps	2 reps	2 reps

4. Electrical and Control Architecture

The ESP32 manages the state machine, WiFi communication, and all I/O. Two PCA9685 PWM boards drive the 18 MG996R servos over I²C, while freeing ESP32 PWM channels. The 12 V drill motor is controlled through a VNH5019 driver, and the 28BYJ-48 seed stepper is driven by a ULN2003 board. All pin assignments are stored in `Config.h`.

Sensors: The HC-SR04 supports closed-loop depth control using `tareDistance()`, which averages five IDLE baseline readings and compares later drilling readings within ± 0.5 cm. The normally-high IR flame sensor is checked every loop and immediately triggers ABORT if activated.

Communication: The ESP32 hosts the `SpiderRobot_102B` WiFi access point. The browser GUI sends commands through `GET /command?cmd=...` and polls `GET /status` every 1.5 s for JSON telemetry.

Figure 2 shows the full circuit diagram.

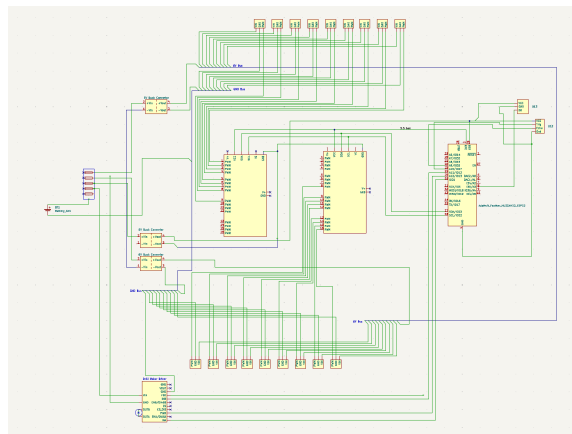


Figure 2: Circuit diagram showing ESP32 GPIO assignments, PCA9685 I²C servo driver connections, L298N drill motor driver, ULN2003 stepper driver, HC-SR04 ultrasonic wiring, IR flame sensor input, and power distribution.

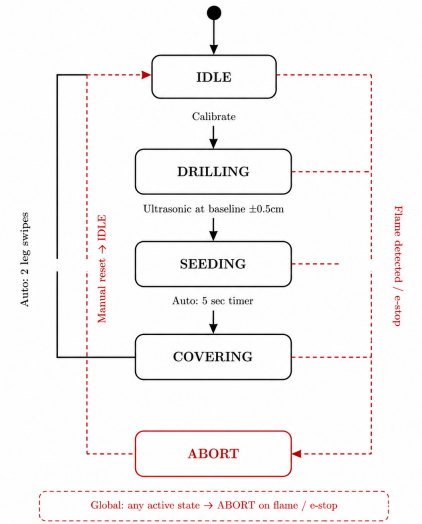
5. State Machine and Event-Driven Logic

The robot's behavior is governed by a finite state machine implemented in `StateMachine.cpp`. Each state has a dedicated service function with a `static bool firstEntry` guard that executes one-time initialization on state entry and ongoing per-loop logic on subsequent calls. `stateMachineLoop()` is called every Arduino `loop()` iteration, ensuring non-blocking operation except during the SEEDING and COVERING phases, which use blocking `delay()` calls for actuator timing precision.

Table 2 describes each state, its entry action, and its exit condition. Figure 3 shows the transition diagram.

Table 2: State machine summary.

State	Entry action	Exit condition
IDLE	Zero all 18 servos; tare ultrasonic baseline (5-sample avg)	Operator sends STATE:DRILLING via GUI
DRILLING	Drill ON; lower body (coxa 90 → 45), then lift; speed scales with distance	Returns to baseline ± 0.5 cm or coxa reaches 90
SEEDING	Drill OFF; stepper rotates 2500 steps	Stepper completes (auto-advance)
COVERING	Right-middle leg sweeps C/F; 2× reps	Animation completes (auto-advance to IDLE)
ABORT	All actuators disabled	Operator sends STATE:IDLE

**Figure 3:** State transition diagram.

6. Function-Critical Decisions and Calculations

6.1. Coxa Servo Selection — Body-Lowering Torque

Each of the six coxa servos must support $\frac{1}{6}$ of the robot's weight through a moment arm equal to femur length $L = 7 \text{ in} = 0.178 \text{ m}$:

$$\tau_{\text{req}} = \frac{mg}{6} \cdot L = \frac{3.205 \times 9.81}{6} \times 0.178 \approx 0.932 \text{ N m} \quad (1)$$

The installed servo is rated at $20 \text{ kg cm} = 1.962 \text{ N m}$, yielding a design safety factor of $SF = 1.962 / 0.932 \approx 2.1$, providing margin for dynamic loads during soil penetration.

6.2. Drill Motor Driver Selection

The Pololu 25D gearmotor operates at 12 V with a rated operating current of $I_{\text{op}} = 0.82 \text{ A}$ (at peak efficiency):

$$P_{\text{in}} = V \cdot I_{\text{op}} = 12 \times 0.82 = 9.84 \text{ W} \quad (2)$$

delivering a maximum mechanical output of 8.9 W at 44% efficiency. Stall current is $I_{\text{stall}} = 5.0 \text{ A}$. The VNH5019 motor driver is rated at 12 A continuous (30 A peak), providing a driver safety factor of $12 / 5.0 = 2.4$ even at stall, well within the thermal envelope for the 60s maximum drill cycle.

6.3. Ultrasonic Depth Sensing — Tare-and-Compare Algorithm

Rather than a fixed threshold, `tareDistance()` samples five readings at IDLE entry to establish a terrain-relative baseline. The DRILLING-to-SEEDING transition fires when:

$$|d_{\text{measured}} - d_{\text{baseline}}| \leq 0.5 \text{ cm} \quad (3)$$

This makes depth control invariant to terrain height, mounting variation, and battery sag, at the cost of a brief initialization delay per cycle.

7. Reflection for Future Students

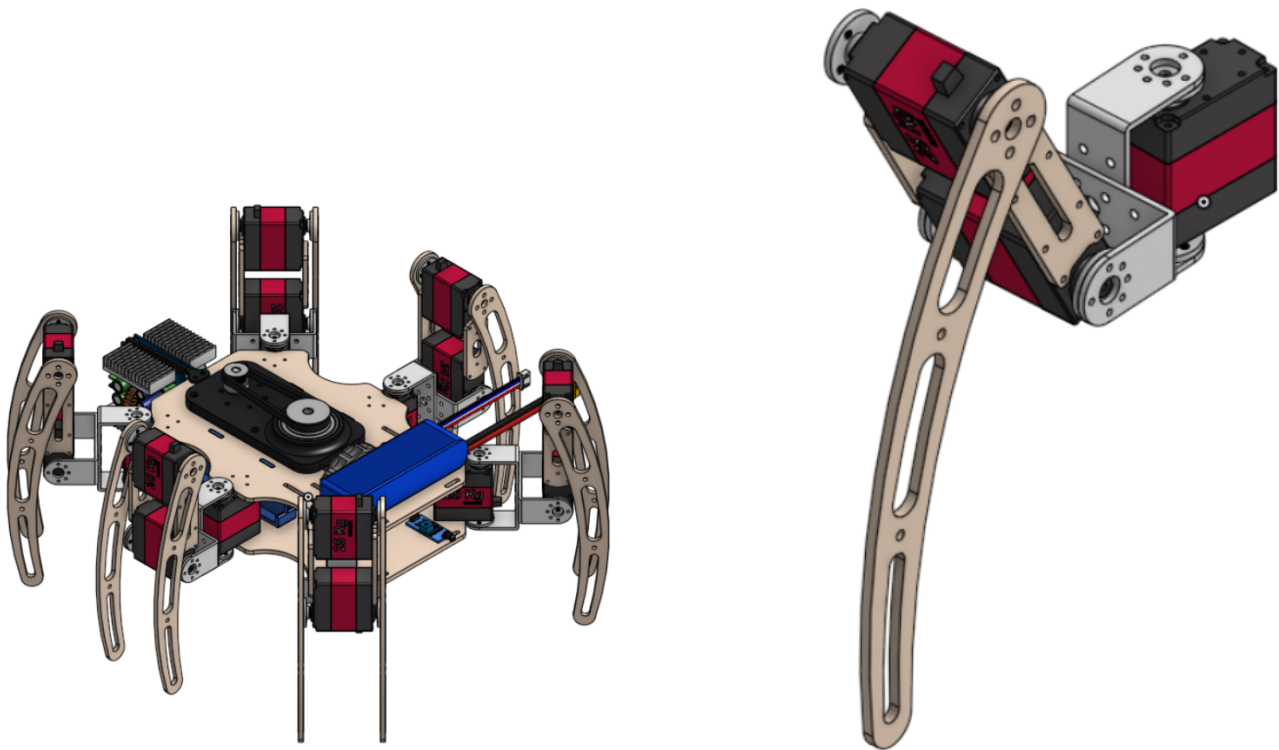
The most useful decision was defining the state machine, WiFi command protocol, and `/status` JSON format before full firmware integration. This gave the team a shared interface and made debugging easier.

Future teams should lock all pin assignments in `Config.h` during initial wiring and build telemetry into the system from the beginning. Adding the telemetry endpoint late required changes across multiple files and introduced a connection-loss bug.

A. Bill of Materials

Component	Description / Supplier	Qty	Unit \$	Total \$
<i>Actuation</i>				
Servo Motor	20 kg·cm, AliExpress	18	10.74	252.13
Gearmotor (Drill)	12 V brushed DC, Pololu	1	56.00	69.90
Stepper Motor	NEMA seed dispenser, Stepper Online	1	20.49	20.49
<i>Electronics / Control</i>				
PWM Expander	PCA9685 16-ch I ² C, Amazon	2	9.99	21.28
Motor Driver	Pololu VNH5019, drill motor	1	12.95	14.21
Logic Converter	Bi-directional 3.3 V/5 V, Amazon	1	4.99	6.29
IR Flame Sensor	Digital, 760–1100 nm, Amazon	1	7.99	9.29
HC-SR04 Ultrasonic	2–400 cm, on hand	1	–	0.00
Capacitors	Servo rail bulk caps, Amazon	1	6.99	8.29
Protoboard	Amazon	2	8.99	19.28
<i>Power</i>				
3S LiPo Battery	5200 mAh, XT60, Amazon	2	–	90.34
5A Buck Converter	5 V output, Amazon	1	14.99	16.29
12V Buck Converter	12 V output, Amazon	1	12.99	14.29
Power Dist. Board	Amazon	1	6.99	8.29
Terminal Block	Amazon	1	14.99	14.99
Grounding Bus	6-position brass terminal, Amazon	1	8.39	9.69
XT60 Connector	Amazon	1	9.89	11.19
30 A Fuse	Amazon	1	4.86	6.16
<i>Mechanical Structure</i>				
Aluminum 6061	Assorted cuts, Jacobs/Amazon	–	–	114.24
Timing Pulley 2GT	2-piece set, Amazon	1	8.99	9.91
Bearing	Amazon	1	16.99	22.03
Drill Brackets	Pololu	1	10.95	12.22
Step Drill Bit	Home Depot	1	8.97	10.17
Coupler (Drill)	Walmart	1	9.99	10.76
Aluminum Shaft	Amazon	1	9.99	11.29
Hardware	M3/M4 screws, McMaster/on hand	–	–	3.11
Sleeve Tube	Amazon	1	7.99	9.29
<i>Wiring / Connectors</i>				
Wire, assorted	14–24 AWG, Amazon	–	–	123.46
Wire Tubing	Amazon	1	10.00	11.30
Heat Shrink Kit	Amazon	1	9.99	11.29
Banana Connectors	Male/female set, Amazon	1	9.99	11.29
Pin Headers	Amazon	1	11.99	13.29
<i>Miscellaneous</i>				
Seeds	Amazon	1	5.41	5.41
Adhesive Pads	Amazon	1	8.99	8.99
Connector Kit	80 pc assorted, Amazon	1	4.99	4.99
Breadboard	On hand	1	–	0.00
			Total	\$993.73

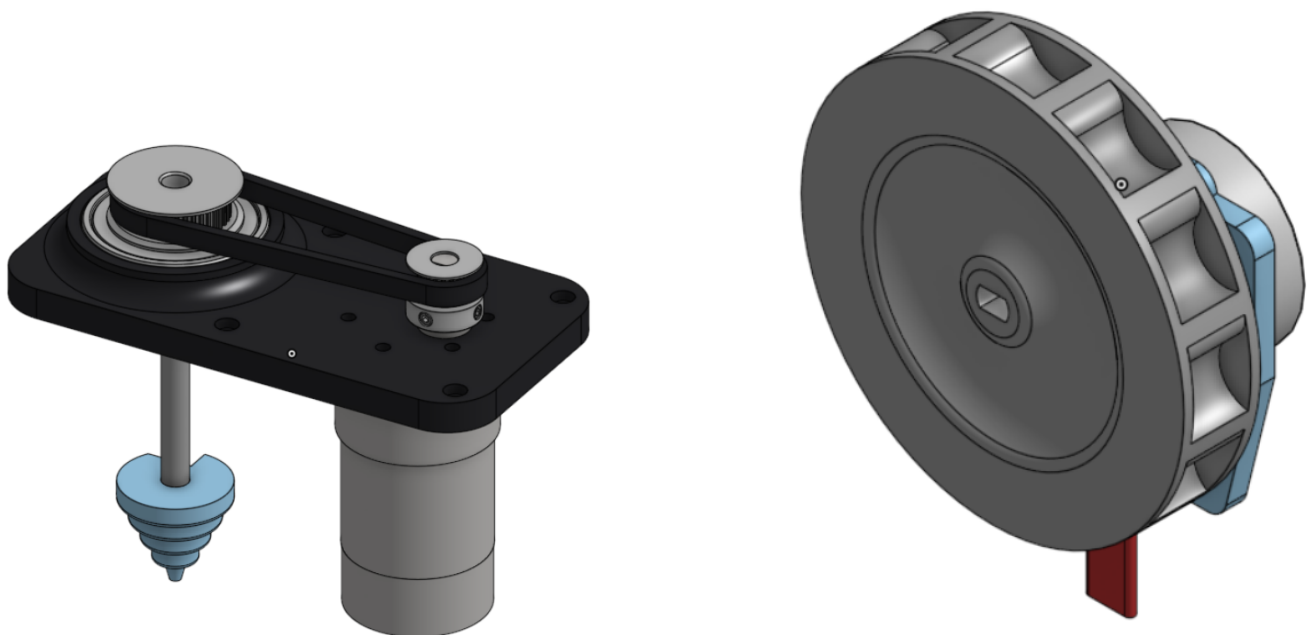
B. CAD Images



(a) Full assembly isometric view.

(b) Single leg transmission detail.

Figure 4: CAD renders showing chassis and leg mechanical transmission elements.



(a) Drill mount and body-lowering mechanism detail.

(b) Seeding meter mechanism.

Figure 5: CAD renders showing the drill mount and seed metering mechanism.